

Notes

Unit 3: *Internal influences on memory*

Lecture 9 (today): *Cognitive Control* - “top-down” influences on learning and memory.

Lecture 10 (tuesday may 12th): *Forgetting* - How, why, when

Lecture 11 (thursday may 14th): *Distortion and False memory*

Lecture 12 (tuesday may 19th): *Emotional memories*

Quiz #3 (thursday may 21st)

Lecture 9: Cognitive Control

PSYCH/COGS 140M 2026.05.07

Cognitive Control

- Cognitive control and Prefrontal Cortex
- Cognitive control and episodic memory

Cognitive Control

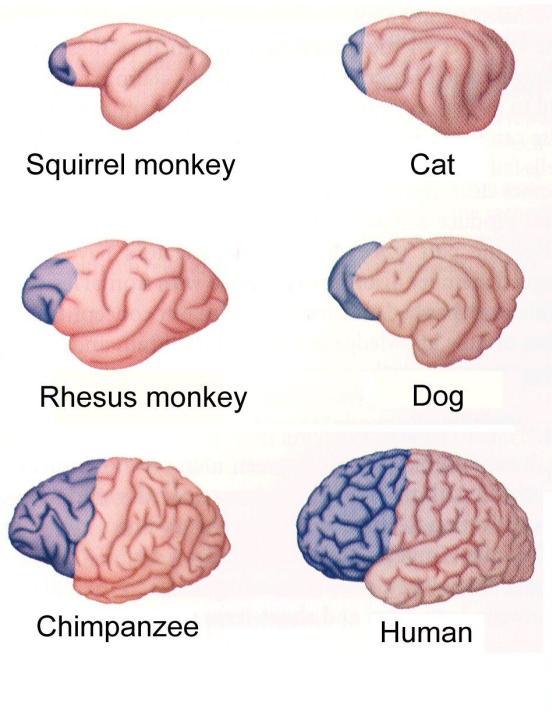
- Cognitive control and Prefrontal Cortex
- Cognitive control and episodic memory

Forms of cognitive control

(we talked about these in the WM lecture!)

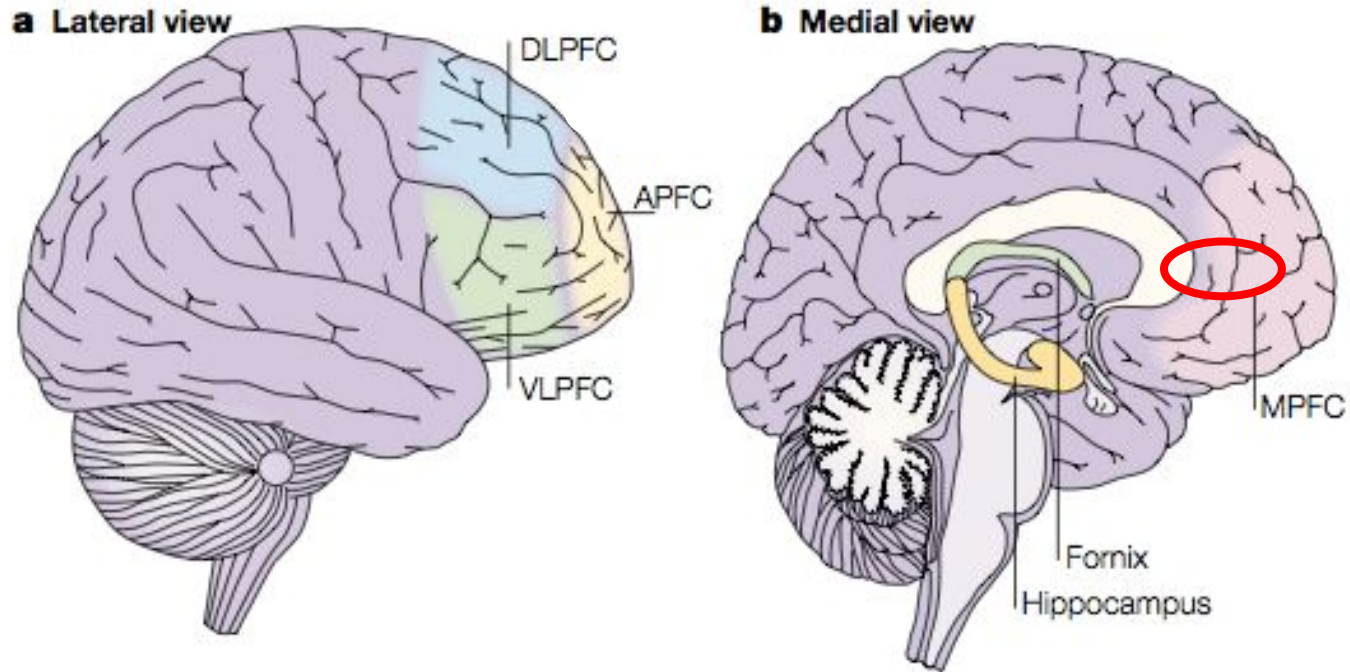
- **Inhibition**: stop a prepotent action, thought, memory
- **Planning**: organizing behaviors to achieve a goal
- **Selection**: choose from competing alternatives
- **Shifting attention**: change focus of attention from one stimulus/thought/task to another
- **Updating**: revise plans/thoughts/memories as behaviors are completed
- **Sequencing**: perform or remember a series of actions/stimuli
- **Monitoring**: evaluate success of behaviors

Prefrontal cortex (PFC)



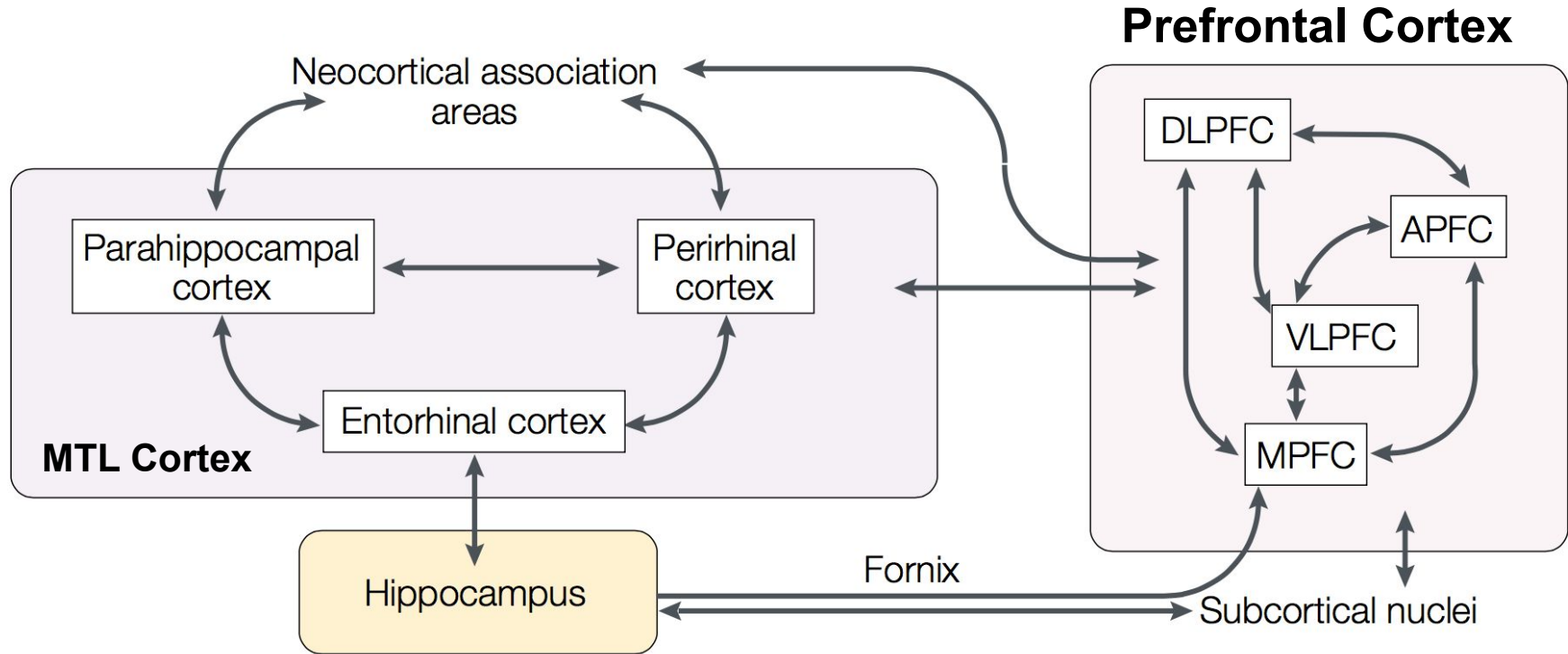
- Anterior section of the frontal lobe
- PFC is proportionally larger in the great apes than in other animals (~1/3 cortex)
- PFC plays a key role in WM and cognitive control

Prefrontal Cortex (PFC)



DLPFC = dorsolateral prefrontal cortex
VLPFC = ventrolateral prefrontal cortex

MTL-PFC Connectivity



Damage to PFC

Patients with PFC lesions show deficits in WM and cognitive control. *Dysexecutive syndrome*

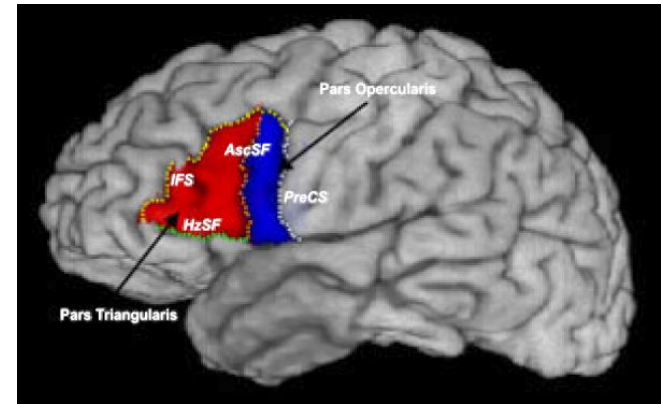
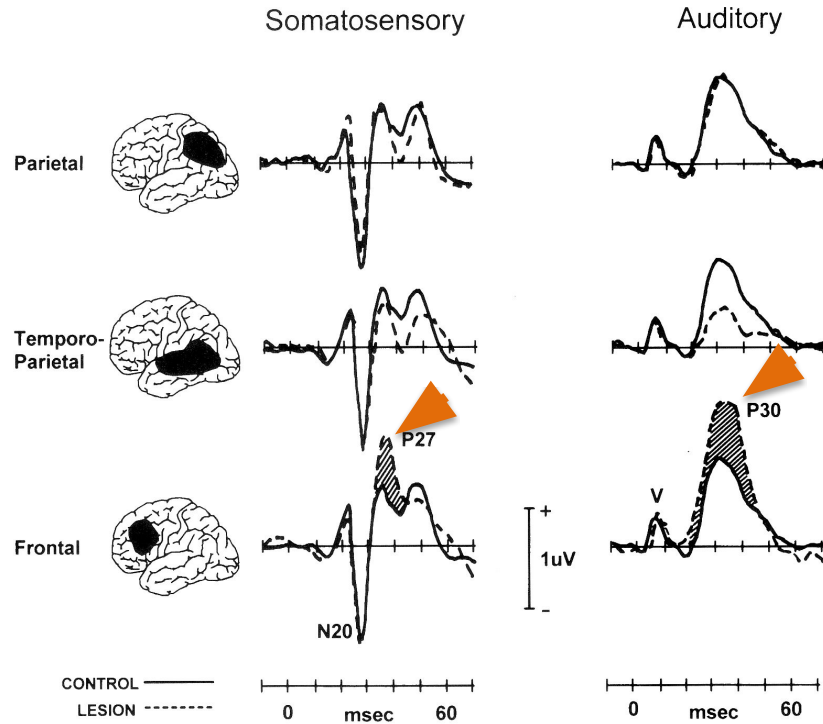
- Increased distractibility
- Perseveration
- Difficulty inhibiting inappropriate responses
- Inability to maintain information given interference
- Difficulty strategizing and planning ahead
- Difficulty initiating behaviors (inactivity)
- Disorganized behaviors
- Responses include irrelevant information
- Failure to recognize or ‘appreciate’ incorrect responses

Damage to PFC

From Bianchi 1922 (pp. 184-185):

“The sight of the window determines the reflex of the jump, but the purpose is now lacking ... Another monkey sees the handle of the door and grasps it, but the mental process stops at the sight of the handle; the animal does not attempt to turn it so as to open the door, but sits on it. Evidently there are lacking all those other images that are necessary for the determination of a series of movements co-ordinated towards one end ... The self-restraint exercised by a normal monkey ... contrasts markedly with the behavior of the mutilated monkey ... ”

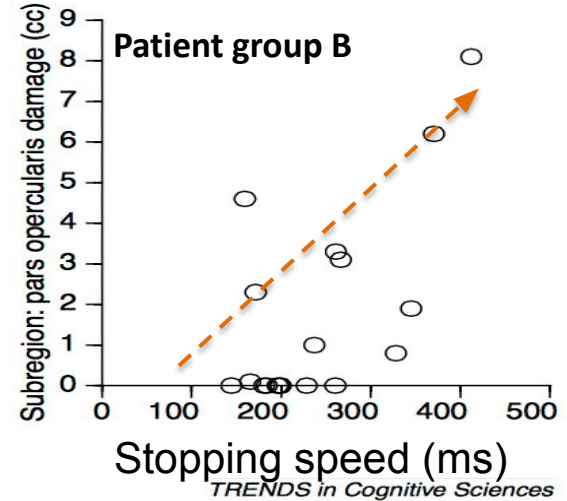
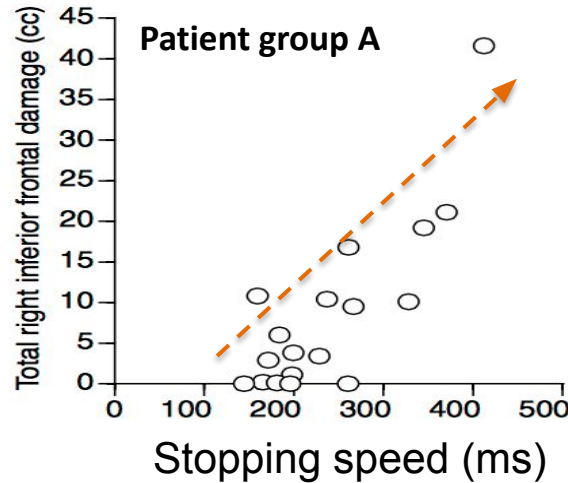
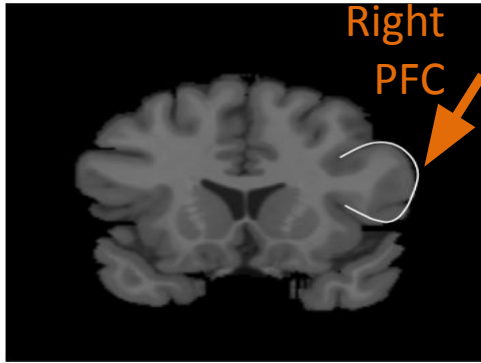
Inhibition



Evoked responses in sensory cortex are *enhanced* by PFC damage

Inhibition

Task: make button presses ASAP unless you are told to stop.



It takes longer to **STOP** a button press with damage to PFC

Cognitive Control

- Cognitive control and Prefrontal Cortex
- Cognitive control and episodic memory

Cognitive Control and Episodic Memory

Damage to prefrontal cortex:

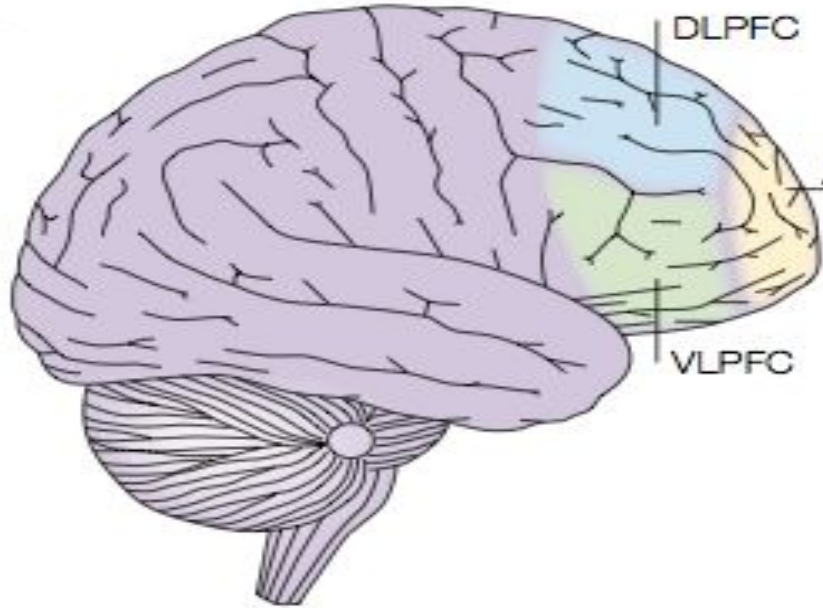
- Does not yield global amnesia
- Does affect episodic memory
 - Produces impairments that generally reflect difficulty ‘controlling’ memory

Cognitive Control and Episodic Memory

Outline

- Relative contributions of DLPFC vs. VLPFC
- Examples of cognitive control during retrieval:
 - Inhibition (stopping) of memory retrieval
 - Selection
 - Source Memory
 - Sequencing

Divisions of Prefrontal Cortex



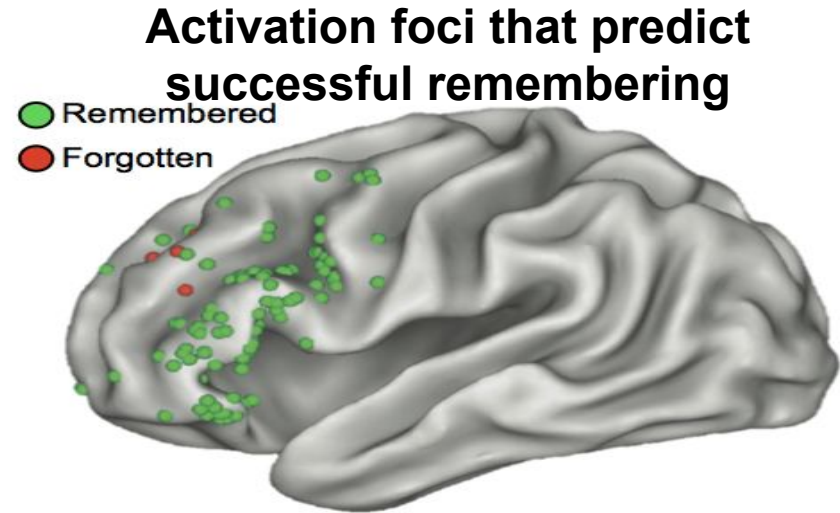
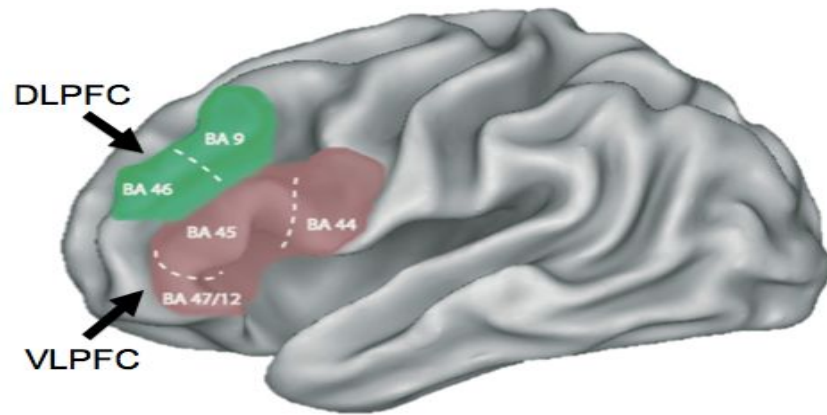
Dorsolateral PFC

- engaged during **recollection-dependent** judgments
- **monitoring of information** in relation to **decision rules**

Ventrolateral PFC

- **cue-directed attention**, maintenance
- selection / **interference resolution**

Control at Encoding: Subsequent Memory Effects



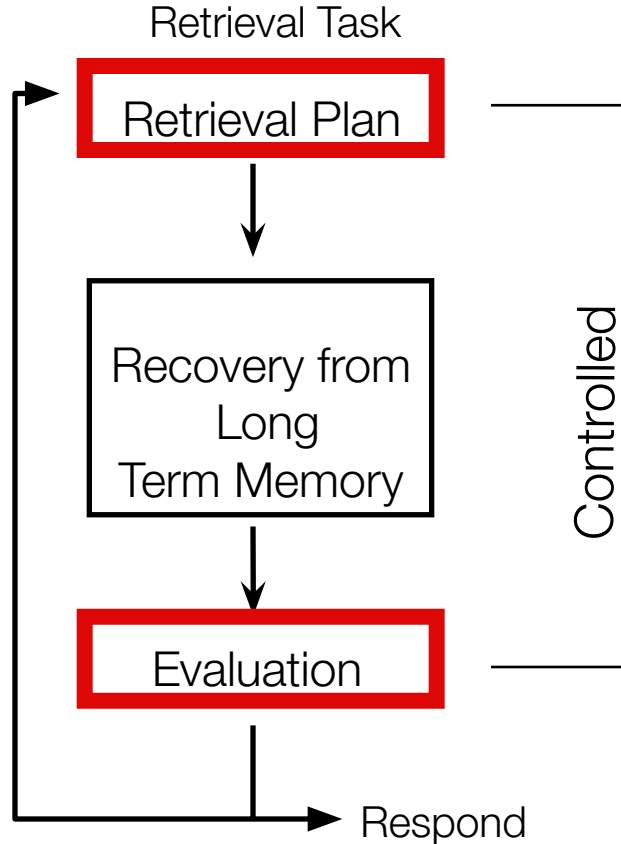
1. Most subsequent memory effects (greater encoding activity for subsequently remembered stimuli) are observed in VLPFC.
2. Effects in DLPFC may reflect demands to relate, reorganize, or manipulate encoded content

Control at Retrieval

At retrieval, control may be required to

- Generate retrieval cues
- Evaluate retrieved information
- Select among competing alternatives
- Or to stop (inhibit) retrieval of unwanted or 'non-target' memories

Component Processes of Retrieval



Pre-Retrieval Processes

- maintain cues in WM
- use cues to bias retrieval

Post-Retrieval Processes

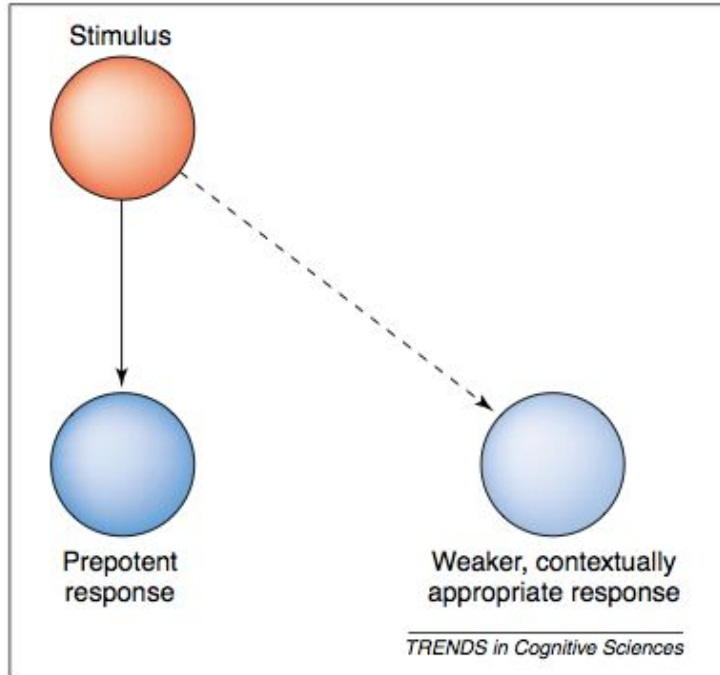
- selection of relevant retrieved details
- decision

Cognitive Control and Episodic Memory

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Inhibition



Can we inhibit memories in the same way we inhibit other behaviors??

Memory Inhibition Demo

Try to remember each of the following pairs of words!

LAWN -
ROACH

JOURNEY -
PANTS

TRUCK-
WRENCH

BOWL-
NAPKIN

Memory Inhibition Demo

LAWN
????

Memory Inhibition Demo

JOURNEY
????

Memory Inhibition Demo

TRUCK
????

Memory Inhibition Demo

BOWL
????

Memory Inhibition Demo

JOURNEY
????

Memory Inhibition Demo

TRUCK
????

Memory Inhibition Demo

LAWN
????

Memory Inhibition Demo

BOWL
????

Memory Inhibition Demo

LAWN
????

Memory Inhibition Demo

JOURNEY
????

Memory Inhibition Demo

TRUCK
????

Memory Inhibition Demo

BOWL
????

Memory Inhibition Demo

JOURNEY
????

Memory Inhibition Demo

LAWN
????

Memory Inhibition Demo

BOWL
????

Memory Inhibition Demo

JOURNEY
????

Memory Inhibition Demo

TRUCK
????

Memory Inhibition Demo



LAWN
????

Memory Inhibition Demo

Now, if you see a word in **GREEN**, try to recall the corresponding word

But if you see a word in **RED**, try to keep the corresponding word out of mind!!

Memory Inhibition Demo

LAWN

????

Memory Inhibition Demo

JOURNEY

????

Memory Inhibition Demo

LAWN

????

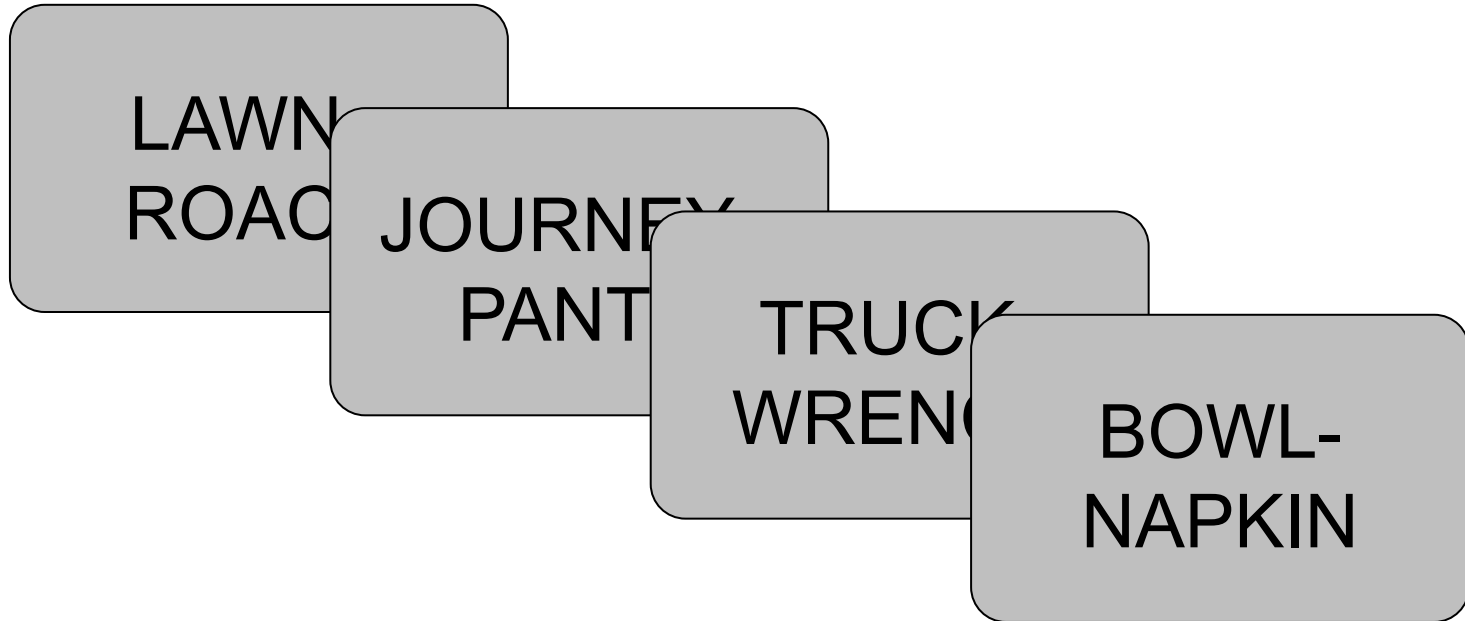
Memory Inhibition Demo

JOURNEY

????

Think/No-Think Task

Learning Phase: Study various word pairs until they are well-learned



Think/No-Think Task

Think / No-Think Phase: For some of the words, try to retrieve the associate; for some of the words try to keep the associate out of mind

LAWN

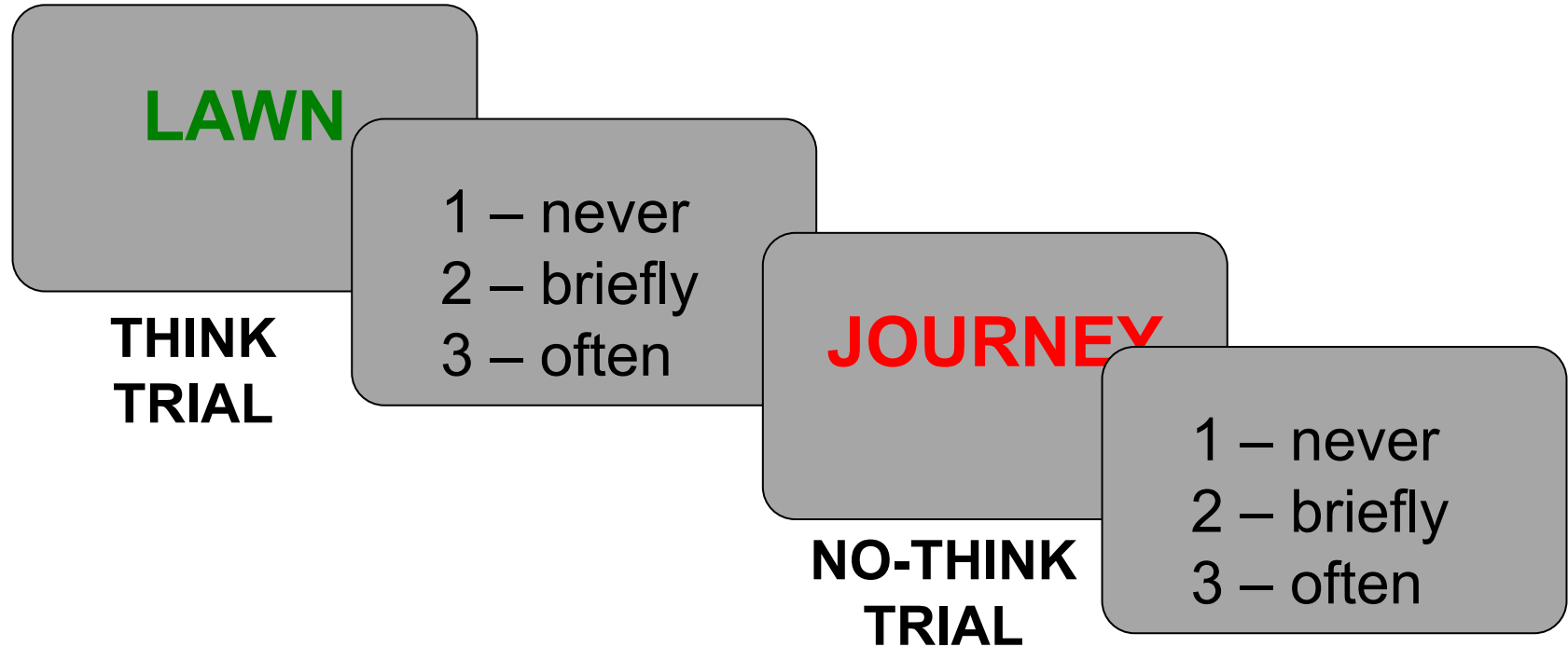
????

JOURNEY

????

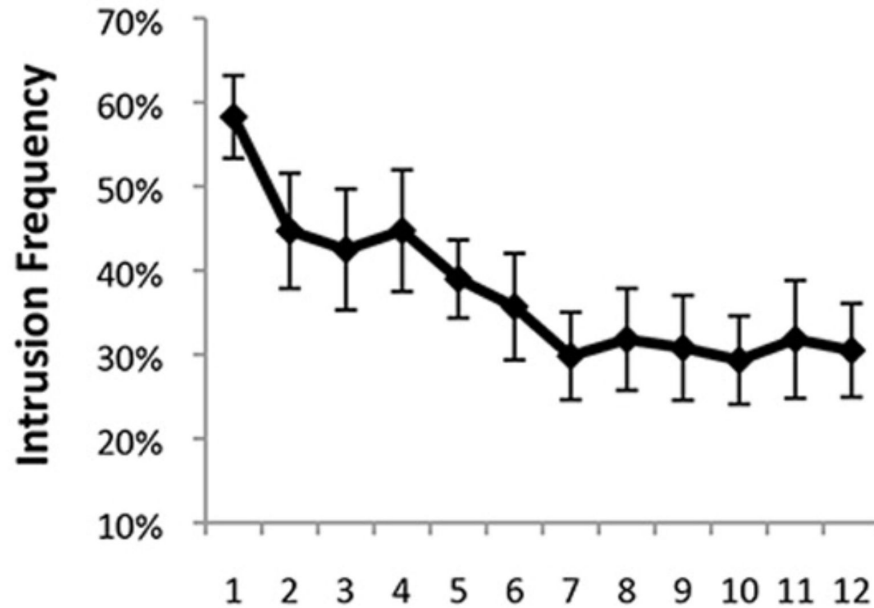
Stopping Memory Retrieval

Think/No-Think task: Think / No-Think Phase

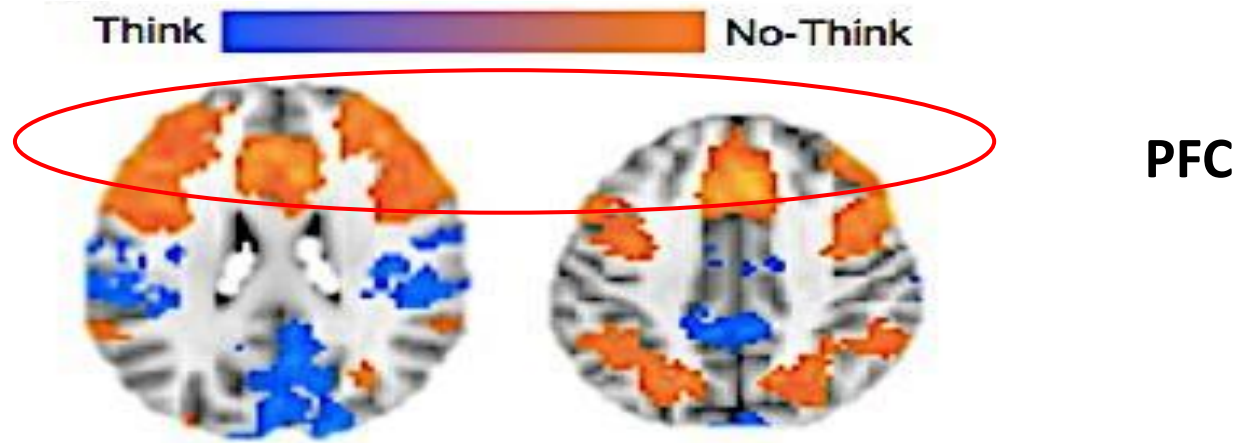


Stopping Memory Retrieval

Intrusions during No-Think Trials

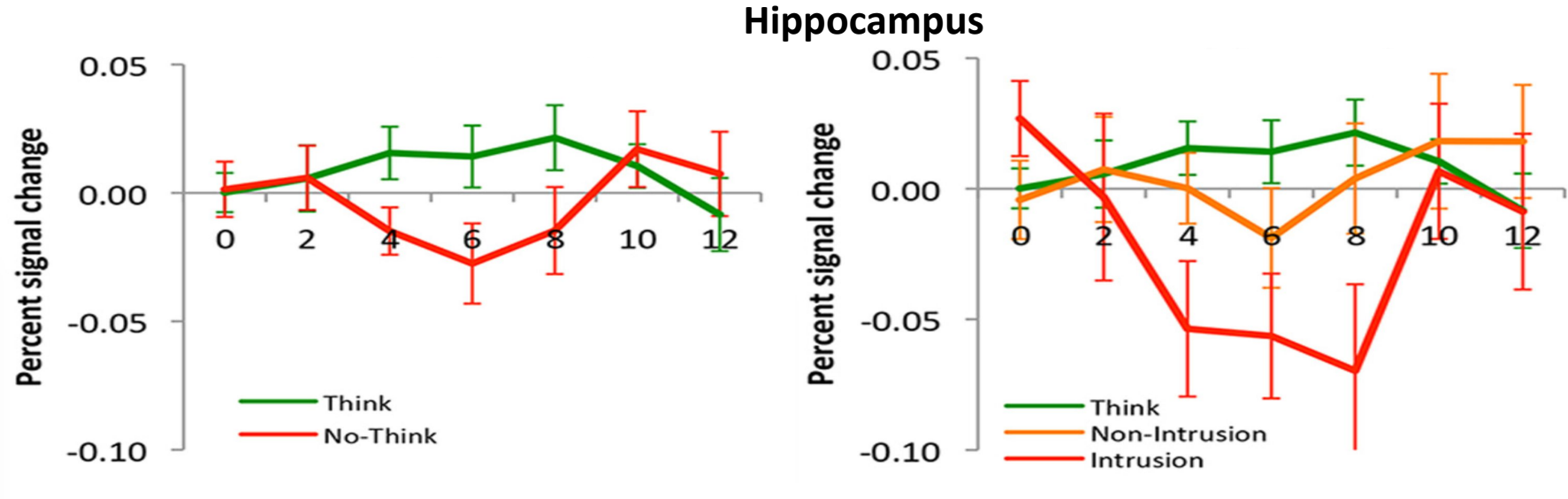


Stopping Memory Retrieval



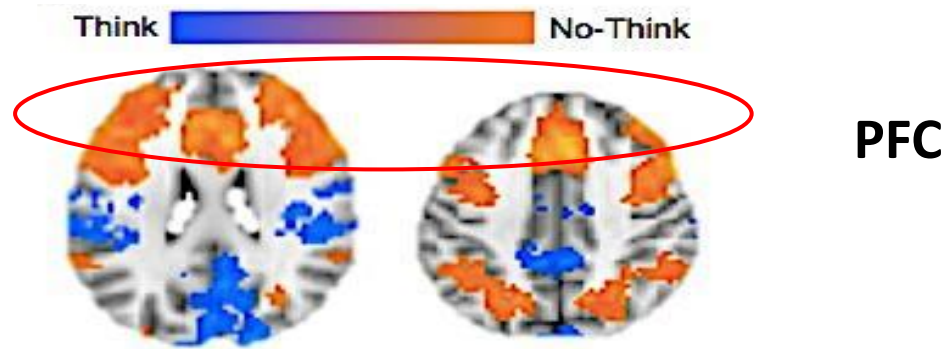
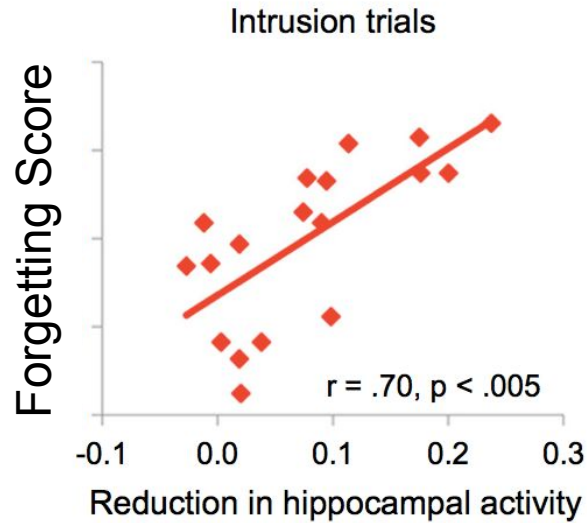
1. Stopping (No-Think) activates prefrontal cortex to a greater degree than retrieving (Think).
2. For well-learned associations, stopping is more effortful than retrieving

Stopping Memory Retrieval



1. No-Think trials are associated with lower hippocampal activation
2. Hippocampal activation is lower when intrusions occur than when they don't occur. Why??

Stopping Memory Retrieval



1. Lower hippocampal activity during intrusion trials correlates with greater forgetting of No-Think items.
2. Together, these results suggest that PFC *reacts* to intrusions by inhibiting the intrusive memories

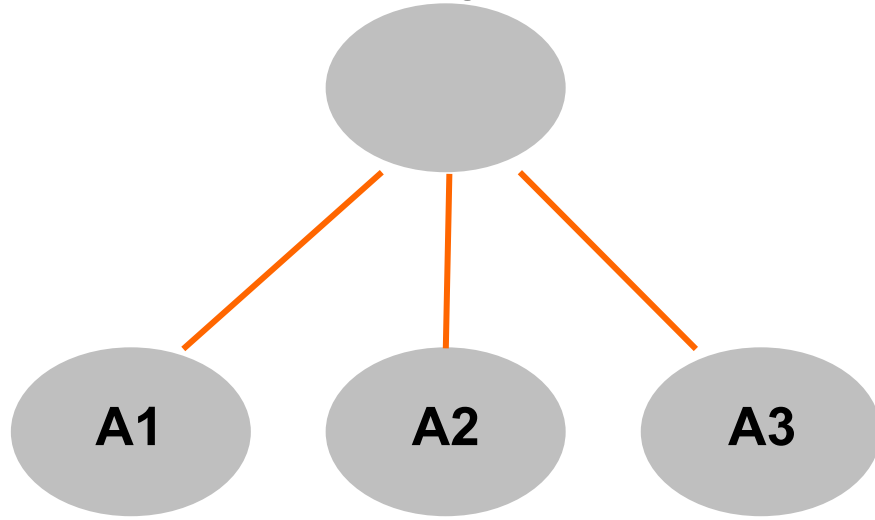
Cognitive Control and Episodic Memory

Outline

- Relative contributions of DLPFC vs. VLPFC
- Examples of cognitive control during retrieval:
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 - Source Memory
 - Sequencing

Selection

Memory Cue



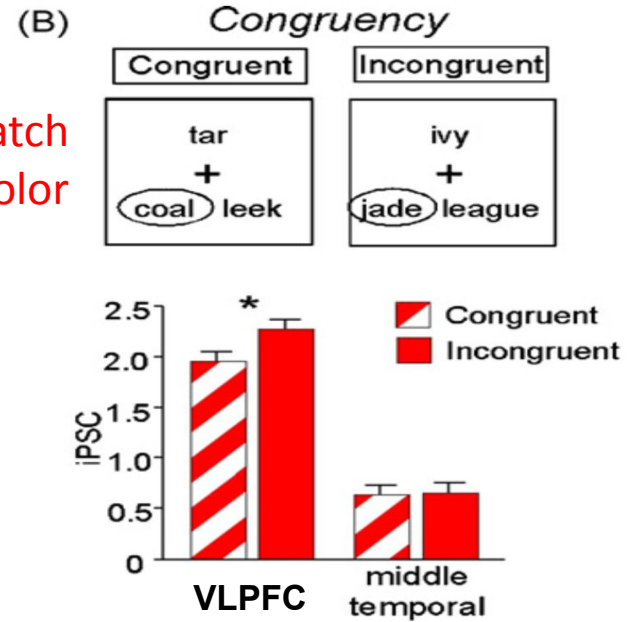
**Goal
relevant**

When multiple memories are active, we 'select' from these competing options

Selection



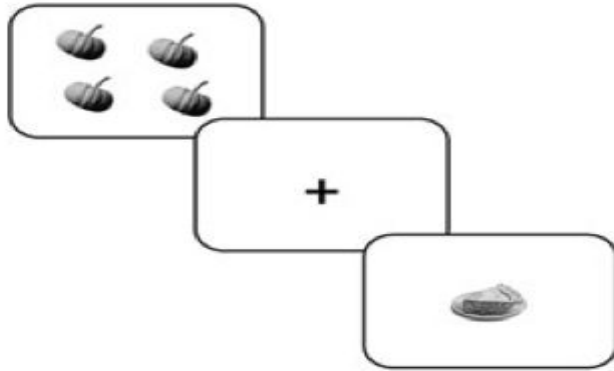
Task: match
by color



Responses against a strong association (incongruent condition) engages VLPFC to a greater degree than responses with a strong association (congruent condition)

Selection

Study session— each image depicts 1, 2, 3 or 4 copies of a common object



answer yes or no

first presentation:

"does one fit inside a lady's shoebox?"

second presentation:

"can you carry all of them at once?"

Test session— hear "pumpkin," then recall and respond 1, 2, 3, 4 or new

while eyes are Shut

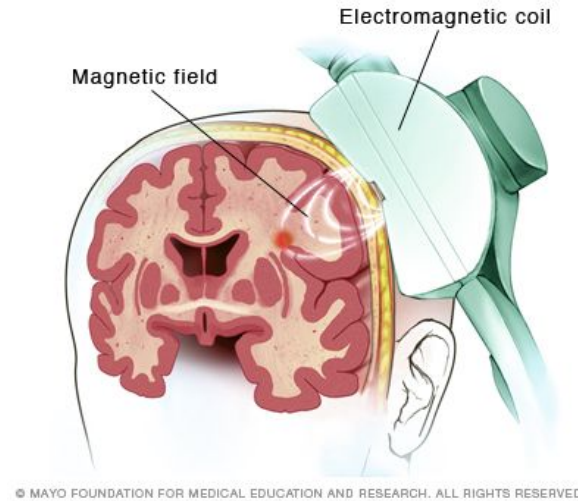


or on Visual Distractor



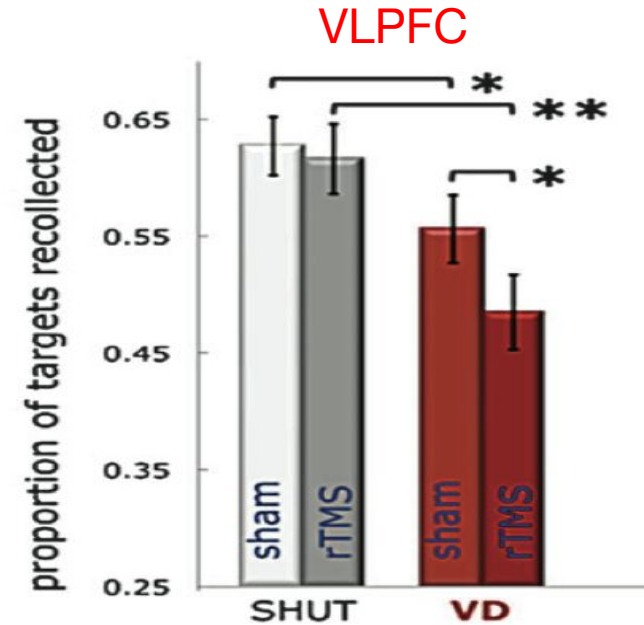
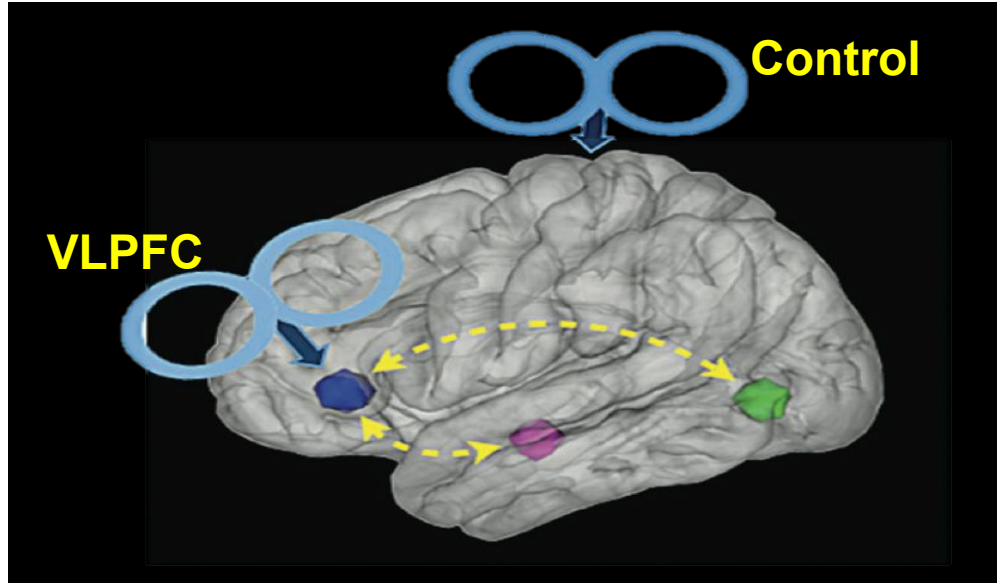
Refresher: Magnetic Neurostimulation

TMS: Transcranial
magnetic stimulation



Selection

TMS applied to either VLPFC or control region.

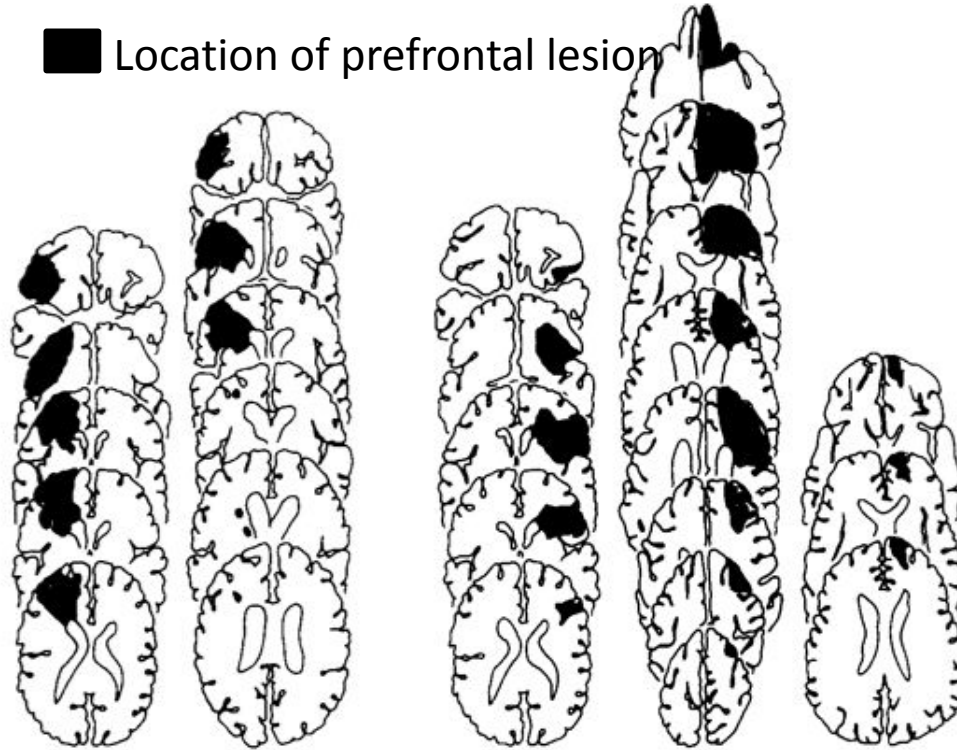


1. Recall was lower with visual distraction (VD; red bars) than with eyes shut (grey bars).
2. TMS to VLPFC exaggerated the effect of distraction.
3. TMS to control region did not influence effect of distraction (not shown).

Source Memory

- Memory for *where* you learned something
- When we remember a piece of information but attribute it to an incorrect source, this is a source error
- E.g., Your friend A tells you a good piece of gossip ... later, you recall this piece of gossip, but you incorrectly ‘remember’ hearing it from B

Source Memory



Source Memory

“Study Phase”

Experimenter

What is the last name of
the author of the Game
of Thrones book series?

Frontal Patient

Sorry, I have no idea.

Experimenter

His last name is Martin.

Source Memory

“Test Phase”

Experimenter

What is the last name of
the author of the
Game of Thrones book?

Frontal Patient

Martin.

Experimenter

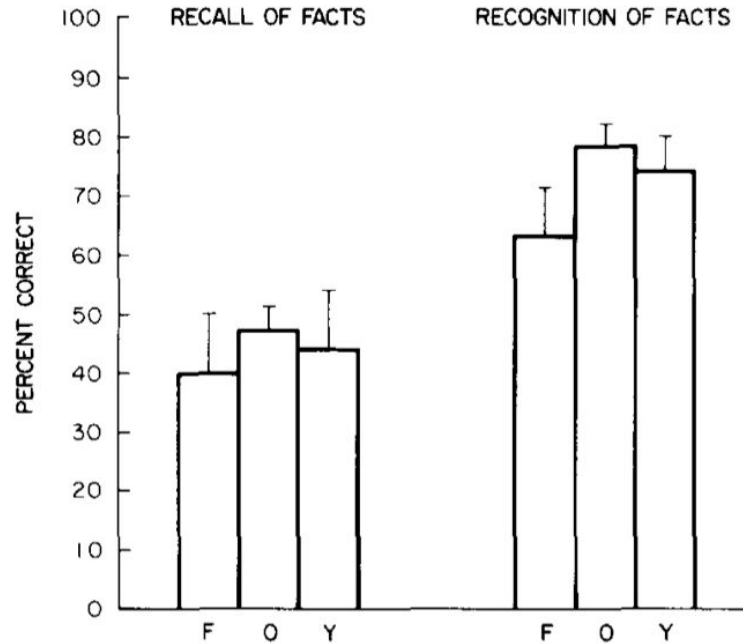
Yes. Where did you
learn that?

Frontal Patient

I read it in a magazine.

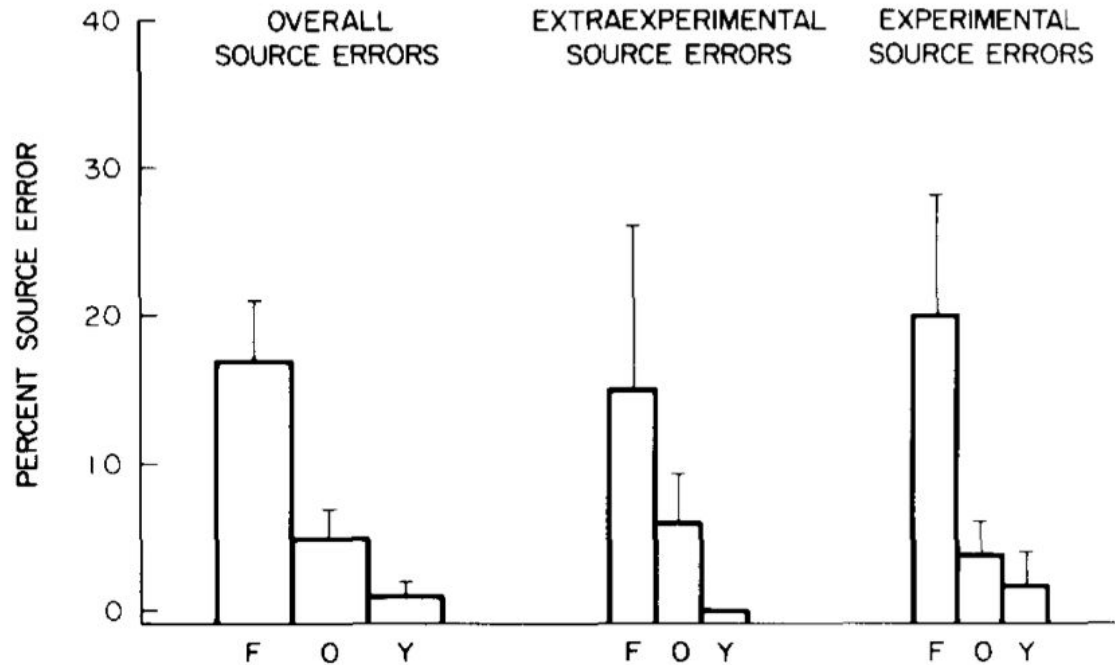
Source Error!

Source Memory



Fact recall and recognition are normal for frontal patients (F) compared to older (O) and young (Y) adults.

Source Memory



Frontal patients make many more source errors,
even when they recognize/recall the fact correctly

Sequencing

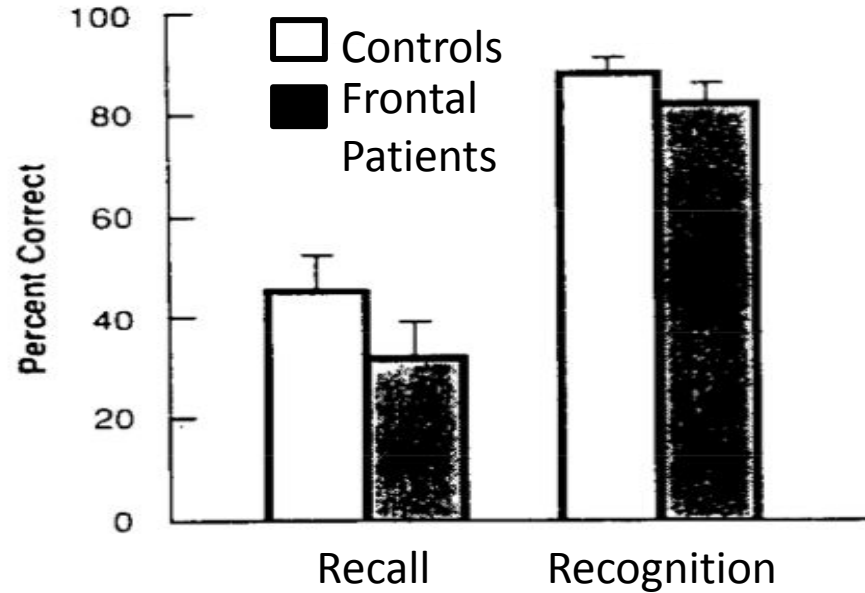
Study List

Candle
Shoe
Phone
Wallet
Pencil
Rake
...

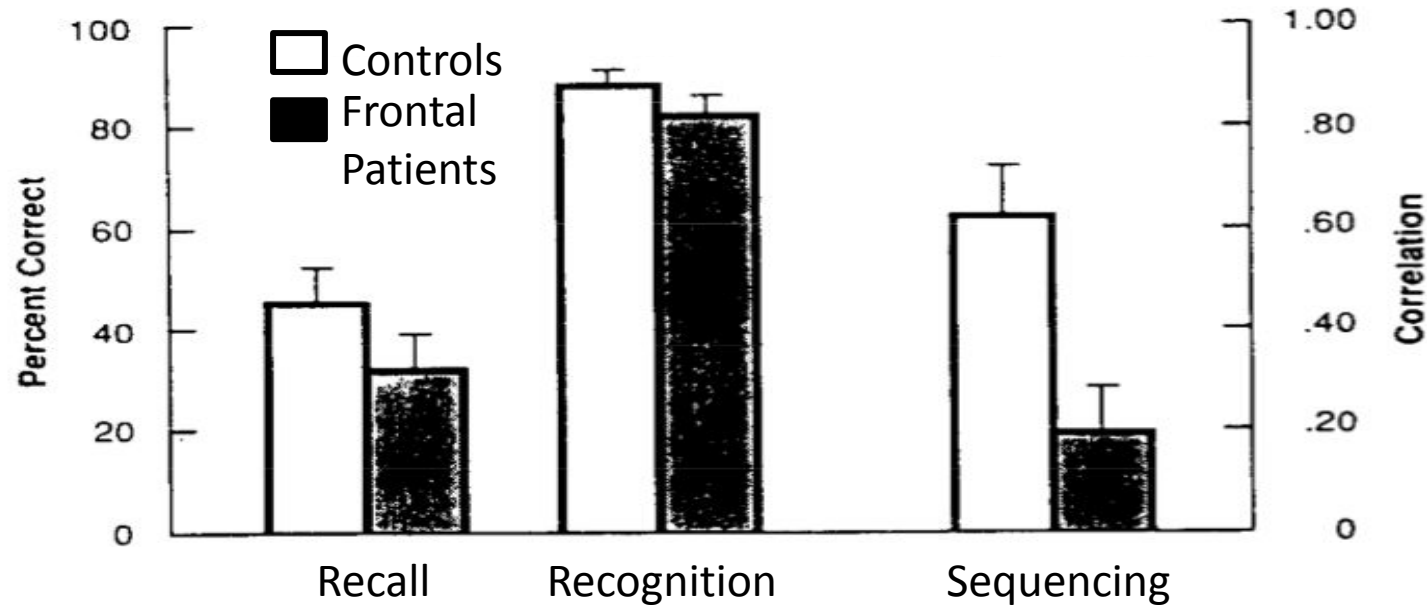
Recall Test

*Try to recall
words in order*

Sequencing Memories



Sequencing Memories



PFC patients show exaggerated impairment at sequencing memories according to their learned order.

Summary: Cognitive Control of Memory

Successful remembering often involves controlling memory

This can include:

- **stopping** recall of a memory
- **selecting** one memory over another memory
- **monitoring** or **evaluating** retrieved memories (as in source memory)
- retrieving the order or **sequence** in which information was learned

Prefrontal Cortex plays a critical role in the cognitive control of memory

Next time...

- Forgetting